

# Homework 3

Math 2031 — Professor Hamilton

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## Section 5.2

### Question 1C

**Statement:**  $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}, n \in \mathbb{N}$

### Proof by Induction:

**Base Case:**  $k = 1, k \in \mathbb{N}$

LHS:  $1^3 = 1$

RHS:  $\frac{1^2(1+1)^2}{4} = \frac{4}{4} = 1$

Since LHS = RHS, the base case is true.

### Induction Step:

Assume  $1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$  is true for  $k$ .

We want to show the statement is true also for  $(k+1)$  by showing the LHS=RHS.

Substituting  $(k+1)$  into the equation:

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Here we notice our IH:

$$\underbrace{1^3 + 2^3 + 3^3 + \dots + k^3}_{\text{Inductive Hypothesis}} + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Replace:

$$\frac{k^2(k+1)^2}{4} + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Factor  $(k+1)^2$  from the LHS:

$$(k+1)^2 \left( \frac{k^2}{4} + k + 1 \right) = \frac{(k+1)^2(k+2)^2}{4}$$

Divide both sides by  $(k+1)^2$ , which is greater than 0.

$$\frac{k^2}{4} + k + 1 = \frac{(k+2)^2}{4}$$

Expanding the LHS we get:

$$\frac{k^2}{4} + k + 1 = \frac{k^2}{4} + k + 1$$

$$\therefore \forall n \in \mathbb{N}, 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

■

### Question 2A

**Statement:**  $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{(2n-1)(2n)(2n+1)}{6}, n \in \mathbb{N}$

### Proof by Induction:

**Base Case:**  $k = 1, k \in \mathbb{N}$

$$\text{LHS: } (2(1) - 1)^2 = 1^2 = 1$$

$$\text{RHS: } \frac{(2(1)-1)(2(1))(2(1)+1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = \frac{6}{6} = 1$$

Since LHS = RHS, the base case is true.

### Induction Step:

Assume  $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{(2k-1)(2k)(2k+1)}{6}$  is true for  $k$ .

We want to show it is also true for  $k+1$  by showing that the LHS=RHS.

Substituting  $k+1$  into our equation:

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6} \text{ is true for } k.$$

Note our IH:

$$\underbrace{1^2 + 3^2 + 5^2 + \dots + (2k-1)^2}_{\text{Inductive Hypothesis}} + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6}.$$

Substitute:

$$\frac{(2k-1)(2k)(2k+1)}{6} + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6}$$

Factoring a  $(2k+1)$  on both sides:

$$(2k+1) \cdot \left( \frac{(2k-1)(2k)}{6} + (2k+1) \right) = (2k+1) \cdot \frac{(2k+2)(2k+3)}{6}$$

Canceling the  $(2k+1)$ , it is greater than 0.

$$\frac{(2k-1)(2k)}{6} + (2k+1) = \frac{(2k+2)(2k+3)}{6}$$

Expanding both sides:

$$\frac{4k^2+10k+6}{6} = \frac{4k^2+10k+6}{6}$$

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$$\therefore \forall n \in \mathbb{N}, 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{(2n-1)(2n)(2n+1)}{6}$$

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### Question 3

**Statement:** For any real number  $a$  except 1,  $1 + a + a^2 + a^3 + \dots + a^n = \frac{(a^{n+1}-1)}{a-1}$

### Proof by Induction:

**Base Case:**  $k = 1, k \in \mathbb{N}$

$$\text{LHS: } 1 + a^1 = a + 1$$

$$\text{RHS: } \frac{a^{1+1}-1}{a-1} = \frac{a^2-1}{1} = \frac{(a+1)(a-1)}{(a-1)} = a + 1$$

Since the LHS = RHS, the base case is true.

**Induction Step:**

Assume  $1 + a + a^2 + a^3 + \dots + a^k = \frac{(a^{k+1}-1)}{a-1}$  is true for  $k$ .

We want to show that it is also true for  $k+1$  by showing that the LHS=RHS.

Substitute  $(k+1)$  into equation:

$$\underbrace{1 + a + a^2 + a^3 + \dots + a^k}_{\text{Induction Hypothesis.}} + a^{k+1} = \frac{(a^{k+2}-1)}{a-1}.$$

Substitute the IH:

$$\frac{(a^{k+1}-1)}{a-1} + a^{k+1} = \frac{(a^{k+2}-1)}{a-1}.$$

Combining the terms on our LHS:

$$\frac{a^{k+1}-1+a^{k+2}-a^{k+1}}{a-1} = \frac{(a^{k+2}-1)}{a-1}.$$

Canceling like terms and rearranging terms:

$$\frac{(a^{k+2}-1)}{a-1} = \frac{(a^{k+2}-1)}{a-1}$$

$$\therefore \forall n \in \mathbb{N} \text{ and } \forall a \in \mathbb{R} : \left( 1 + a + a^2 + a^3 + \dots + a^n = \frac{(a^{n+1}-1)}{a-1} \right)$$

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**Question 8**

**Statement:** “The Fibonnaci numbers  $f_n, n = 1, 2, 3, \dots$  are defined recursively by the formulas  $f_1 = 1, f_2 = 1, f_n = f_{n-1} + f_{n-2}$  for all  $n \geq 3$ .”

- (A)  $f_1 = 1, f_2 = 1, f_3 = 2, f_4 = 3, f_5 = 5, f_6 = 8, f_7 = 13, f_8 = 21, f_9 = 34, f_{10} = 55$   
 (B)  $f_1 + f_2 = 2, f_1 + f_2 + f_3 = 4, f_1 + f_2 + f_3 + f_4 = 7$   
 (C) **Statement:**  $f_1 + f_2 + f_3 + \dots + f_n = f_{n+2} - 1, n \geq 3$

**Proof by Induction:**

**Base Cases:**  $i = 1$

LHS:  $f_1 = 1$

RHS:  $f_3 - 1 = 2 - 1 = 1$

Since the LHS=RHS, the statement is true for  $i = 1$ .

**Base Cases:**  $i = 2$

LHS:  $f_1 + f_2 = 1 + 1 = 2$

RHS:  $f_4 - 1 = 3 - 1 = 2$

Since the LHS=RHS, the statement is true for  $i = 2$ .

**Induction Step: Fix for  $k > 3$**

Let the equation  $f_1 + f_2 + f_3 + \dots + f_i = f_{i+2} - 1$  be true for  $1 \leq i < k$ .

We want to show the statement is true for  $(k)$  by demonstrating the LHS=RHS.

Substituting  $(k)$ :

$$\underbrace{f_1 + f_2 + f_3 + \dots + f_{k-1}}_{\text{Induction Hypothesis.}} + f_k = f_{k+2} - 1$$

Replace the IH as  $k - 1$  is within our assumption range:

$$f_{k+1} - 1 + f_k = f_{k+2} - 1.$$

Add 1 to both sides:

$$f_{k+1} + f_k = f_{k+2}$$

By the definition of the Fibonacci sequence:

$$f_{k+2} = f_{k+1} + f_k$$

Thus:

$$f_{k+1} + f_k = f_{k+1} + f_k$$

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$$\therefore f_1 + f_2 + f_3 + \dots + f_n = f_{n+2} - 1, n \geq 3$$

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## Additional Problems:

### Question 1

**Statement:** Prove that for every natural number  $n$ ,  $3^n \geq 1 + 2^n$

### Proof by Induction:

**Base Case:**  $k = 1$

$$\text{LHS} = 3^1 = 3$$

$$\text{RHS} = 1 + 2^1 = 1 + 2 = 3$$

Since the LHS  $\geq$  RHS the statement is true for  $k = 1$ .

### Induction Step:

Let the statement  $3^k \geq 2^k + 1$  be true for  $k$ .

We want to prove the statement is true for  $(k + 1)$  by showing that the LHS  $\geq$  RHS.

Substituting:

$$3^{k+1} \geq 2^{k+1} + 1$$

Rewrite:

$$3 \cdot \underbrace{3^k}_{\text{Inductive Hypothesis}} \geq 2 \cdot 2^k + 1$$

Replace with IH:

$$3 \cdot (2^k + 1) \geq 2 \cdot 2^k + 1$$

Simplify further:

$$3 \cdot 2^k + 3 \geq 2 \cdot 2^k + 1$$

Subtract 1 from both sides:

$$3 \cdot 2^k + 2 \geq 2 \cdot 2^k$$

Finalize:

$$3 \cdot 2^{k+1} \geq 2^{k+1}$$

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$\therefore \forall n, 3^n \geq 1 + 2^n$  since we have shown that the LHS  $\geq$  RHS.

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## Question 2

**Statement:** Prove that for every number natural number  $n$ ,  $n^3 + 5n + 6$  is divisible by 3.

### Proof by Induction:

#### Base Case: $k = 1$

We want to show that our result is an integer multiple of 3.

$$1^3 + 5(1) + 6 = 6 + 6 = 12$$

$$12 = 4(3)$$

$\therefore$  the statement is true for  $k = 1$ .

#### Induction Step:

Let the statement  $k^3 + 5k + 6 = 3b$  be true for  $k$ , where  $k$  and  $b$  are integers.

We want to show that the statement is true for  $(k + 1)$  by showing it is also divisible by 3.

Substituting LHS:

$$(k + 1)^3 + 5(k + 1) + 6$$

Expanding further:

$$k^3 + 3k^2 + 3k + 1 + 5k + 5 + 6$$

Simplifying:

$$\underbrace{k^3 + 5k + 6}_{\text{Inductive Hypothesis}} + 3k^2 + 3k + 6$$

Inductive Hypothesis

Replace the IH:

$$3b + 3k^2 + 3k + 6$$

Factor a 3:

$$3(b + k^2 + k + 2)$$

Since  $(k + 1)^3 + 5(k + 1) + 6$  is an integer multiple of 3, it must also mean it is divisible by 3.

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$\therefore \forall n \in \mathbb{N}, n^3 + 5n + 6$  is divisible by 3.

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## Question 3

**Statement:** Let  $a_1 = 2, a_2 = 4$  and  $a_n = 5a_{n-1} - 6a_{n-2}$ , for  $n \geq 3$ .

Prove that for every natural number  $n$ ,  $a_n = 2^n$ .

### Proof by Induction:

#### Base Cases: $i = 1$

LHS:  $a_1 = 2$  RHS:  $2^1 = 2$  Since the LHS=RHS the statement is true for  $i = 1$ .

**Base Cases:  $i = 2$** 

LHS:  $a_2 = 4$  RHS:  $2^2 = 4$  Since the LHS=RHS the statement is true for  $i = 2$ .

**Induction Step:**

Assume the statement  $a_n = 2^n$  for the above defined sequence is true for  $1 \leq i < k$ , fix  $k > 3$ .

We want to show that the statement is true for  $a_k = 2^k$  as well by showing the LHS=RHS.

By definition:

$$a_k = 5a_{k-1} - 6a_{k-2}$$

Note here that  $(k-1)$  and  $(k-2)$  both fall within our assumption range above, thus:

$$a_{k-1} = 2^{k-1} \text{ and } a_{k-2} = 2^{k-2}$$

Rewriting our expression:

$$5(2^{k-1}) - 6(2^{k-2}) = 2^k$$

Note here that  $6 = 3 \cdot 2$  thus we can rewrite as:

$$5(2^{k-1}) - 3(2^{k-1}) = 2^k$$

Factor out a  $2^{k-1}$ :

$$2^{k-1}(5 - 3) = 2^k$$

Simplify:

$$2^{k-1}(2) = 2^k$$

Finally:

$$2^k = 2^k$$

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$\therefore$  for the sequence above,  $\forall n \in \mathbb{N}, a_n = 2^n$ .

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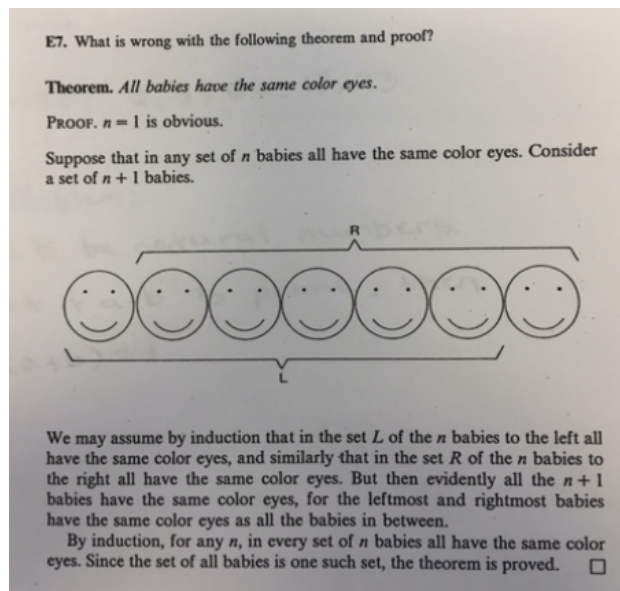
**Question 4**

Figure 1: Babies Problem

The main issue with this proof lies in the fact that it makes use of strong-form induction in its argumentation whilst using the logic of a regular induction proof. Note how the author uses the left side of the set  $L$  in order to accommodate for the end, however the base case only allows us to assume  $n$  is true, not for a range of babies in  $L$ .